

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 04/06/18

Day :Monday

Time: 10.00 am to 10.30am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : MPT- June 2018
Semester: 3
Section : I
Course Code : PT911
Course Name: Motor control, motor development and motor learning

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs) (Encircle the correct answer)

(10 × 1 = 10)

1. Which is not true concerning the "degrees of freedom" problem?
 - a. it arises when a complex system is needed to produce a specific result
 - b. it is important to know which elements must be controlled
 - c. the degrees of freedom reflect the number of independent elements of the system
 - d. It refers only to the degrees of freedom available in human joints
2. The central nervous system (CNS) consists of the
 - a. nerves that connect the spinal cord with other parts of the body
 - b. brain and spinal cord
 - c. cerebellum and thalamus
 - d. nerves of the sensory system

3. Which part of the neuron contains the nucleus and works to maintain homeostasis of the neuron?
 - a. Axon
 - b. Dendrites
 - c. Soma (or cell body)
 - d. interneuron
4. Alpha motor neurons connect with
 - a. golgi tendon organs
 - b. muscle spindles
 - c. sensory receptors
 - d. skeletal muscle
5. What is the resting state of the neuron called?
 - a. Polarisation
 - b. Depolarisation
 - c. Action potential
 - d. Neural transmission
6. According to Gentile's two-dimensional skill classification system, the most complex skills would involve
 - a. body transport, object manipulation, regulatory conditions in motion, and intertrial variability
 - b. body stability, no object manipulation, stationary regulatory conditions, and no intertrial variability
 - c. body transport, no object manipulation, regulatory conditions in motion, and no intertrial variability
 - d. body stability, object manipulation, stationary regulatory conditions, and intertrial variability
7. Performance changes tend to be largest in which stage of learning?
 - a. Cognitive
 - b. Associative
 - c. Automatic
 - d. Gross
8. Retention tests are generally used to measure which performance characteristic of motor skill learning?
 - a. Improvement
 - b. Persistence
 - c. Stability
 - d. Adaptation
9. Which of the following would best fit the definition of a motor skill?
 - a. Strength
 - b. Flexibility
 - c. Walking
 - d. Balance
10. In Gentile's two-dimensional classification system, the environmental context dimension has which two characteristics?
 - a. Body transport and object manipulation
 - b. Regulatory conditions and intertrial variability
 - c. Body transport and intertrial variability
 - d. Regulatory conditions and object manipulation

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer)

True False

11. An open-loop control system involves feedback
12. Junction between two neurons is known as neuromuscular junction
13. Interneurons sends information within nervous system
14. A neuron can partially fire an action potential to indicate different stimulus intensities
15. A monosynaptic reflex is faster than polysynaptic reflex

☐	☐
☐	☐
☐	☐
☐	☐
☐	☐

Match the following

(5 × 1 = 5)

16.	Area involved in planning movement and organizing the coordination of movement	A. Basal ganglia B. Cerebellum C. Thalamus D. Premotor area (PMA) E. Parietal cortex F. Occipital cortex G. Medulla
17.	Error detection and correction	
18.	Relay station of sensory information	
19.	Area where sensory and motor pathways cross and merge	
20.	areas that are vital for perception of sensory information	

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of M.P.T Examination

June 2018

PT 911 Motor control, motor development and motor learning

Date: 04/06/18

Time: 10.30 am to 1.00 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Define synergy
- b. Define motor control
- c. List righting and equilibrium reactions
- d. Distinguish learning and performance
- e. Illustrate two key functions of feedback in learning a skill
- f. Significance of primitive reflex testing
- g. What is Bernstein's problem?

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Briefly discuss basic tenets of reflex maturation theory of motor development
- b. Describe three major limitations of reflex theory of motor control
- c. Compare Schmidt's schema theory and Adam's closed loop theory
- d. Describe any five types of feedback
- e. Discuss the influence of various practice conditions on learning

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. Discuss in detail about dynamic systems theory of motor control
- b. Discuss in detail about development of walking based on reflex maturation theory
- c. Define motor learning. Discuss various stage of motor learning and add a note on its clinical implications